

Class 2

April 13, 2026

Today

- Monopoly and Trading Game Recap and Questions
- Production Game Set-up
 - Initial Parameters
 - Cobb Douglas Production Function
 - Little's Law
- Accounting
 - Financial Statements
 - Contribution Margin
 - Break-even
- Build Products
 - Research and Development
 - Pricing
 - Trade
- Production Game Play
 - 1 hour in class (2 quarters)
 - 2 hours from 7 - 9 (4 quarters, work on homework concurrently, monitor game as needed after 830)

Important Links and Documents for Today

Production game link is on Canvas home page and [here](#).

DOCUMENTS

- Gleacher Game Case Brief – Canvas
- Production Model for the Bench Factory - Canvas
- Production Game Market Research – Canvas
- Gleacher Game Research and Design Guide – Canvas
- Gleacher Game Player Manual – Canvas and one copy per team handed out in class
- Quick Reference Guide for Production Game – Canvas and handed out to you in class
- Gleach Game Suggested Team Roles - Canvas

Integrated Strategic Management

Creating Value in Complex
and Ambiguous Settings

Monopoly Game Results

Monopoly Game – Why?

- Pricing
- Demand
- Reorder Points
- Interface

Relevant Formulas for Monopoly Game

Demand Function	$Q^d = a - bP$
Demand Curve	$P = \frac{a}{b} - \frac{1}{b}Q^d$
Max WTP	$\text{Max WTP} = \frac{a}{b}$
Optimal Price	$P = \frac{\text{Max WTP} + \text{Marginal Cost}}{2}$
Expected Daily Demand	$E(Q_{\text{daily}}^d) = np = \text{Daily Arrivals} * \text{Probability of Buying}$
Standard Deviation of Daily Demand	$S(Q_{\text{daily}}^d) = \sqrt{np(1-p)}$
Cost of Overstocking	$C_o = \text{Materials Cost} * \text{Annual Interest Rate}$
Cost of Understocking	$C_u = \text{Contribution Margin per Unit} * \text{\# of Inventory Cycles per Year}$ $C_u = (\text{Price} - \text{Materials Cost}) * \frac{\text{Expected Daily Sales} * \text{Days per Year}}{\text{\# of Units per Inventory Cycle (Batch Size)}}$
Critical Fractile	$\text{Critical Fractile} = \frac{C_u}{(C_u + C_o)}$
Optimal Reorder Point	$\text{Reorder Point} = (w * \mu_d) + (z * \sigma_d * \sqrt{w}) = \text{Lead Time Demand} + \text{Safety Stock},$ where w is lead time, μ_d is average daily demand, z is z-score, σ_d is standard deviation of demand

Monopoly Game Results

Rank	World	Team Name	Cash Position
1		Zion	\$ 1,435,313
2		Gotham	\$ 1,421,300
3		Globex	\$ 1,419,961
4		Dune	\$ 1,414,372
5		Vulcan	\$ 1,408,154
6		Panem	\$ 1,385,201
7		B612	\$ 1,362,175
8		Westeros	\$ 1,283,652

WINNER!!

Max WTP

Hormone = \$500

Specialty = \$1,000

Optimal Prices

Hormone = \$300

Specialty = \$550

Expected Daily Demand

Hormone = 12

Specialty = 6.3

Optimal Reorder Points

Hormone = 57.31

Specialty = 32.85

Excess Capacity

21.7 units per day

Let's trade!

Integrated Strategic Management

Creating Value in Complex
and Ambiguous Settings

Trading Game Results

Trading Game – Why?

- Pricing
 - Pricing for retail and wholesale customers
 - Double marginalization problem
- Supply Chain Risks
 - Demand side
 - Supply side
- Capacity
 - Excess capacity
- Shipping Agreement Terms
- Shipping
 - Mail vs. container
- Stockouts
- Reorder Points
- Reports

Scoreboard

Rank	World	Team Name	Cash Position
1		Serenity	<u>\$ 2,728,983</u>
2		Metropolis	<u>\$ 2,644,542</u>
3		Vulcan WINNER!!	<u>\$ 2,543,009</u>
4		Dune	<u>\$ 2,399,097</u>
5		Westeros	<u>\$ 2,326,490</u>
6		Panem	<u>\$ 2,169,325</u>
7		Gotham	<u>\$ 1,962,973</u>
8		Zion	<u>\$ 1,958,344</u>
9		Globex	<u>\$ 1,756,041</u>
10		B612	<u>\$ 1,467,302</u>

Trading Game Results

Market Grid: Unit Sales and Revenue

	B612	Dune	Globex	Gotham	Metropolis	Panem	Serenity	Vulcan	Westeros	Zion	Total
Hormone	3,698	3,712	2,600	2,413	3,791	3,891	3,782	3,786	3,745	3,376	34,794
Circulation	2,434	60									2,494
Cardiac	140	2,434	360	286	440	370	791	488	397	520	6,226
Cancer		318	2,421	245		319	49	239	269	328	4,188
Gas	172	257	175	2,929	562	330	355	380	696	470	6,326
Thyroid	164	1,140			2,430	334	1,455		974		6,497
Opiate	296	302	162	365	840	458	88	2,208	753	520	5,992
Motion		205	315	290	177	2,303	611	370	197	795	5,263
Proteins		763	185		1,300	889	2,458	210	562	837	7,204
Botulinum	358	914	130	436		180	1,230	697	1,970	400	6,315
Amphetamine			130	330		255				2,580	3,295
Total	7,262	10,105	6,478	7,294	9,540	9,329	10,819	8,378	9,563	9,826	88,594

	B612	Dune	Globex	Gotham	Metropolis	Panem	Serenity	Vulcan	Westeros	Zion	Total
Hormone	\$ 1,165,200	\$ 1,167,400	\$ 832,500	\$ 778,100	\$ 1,188,600	\$ 1,176,400	\$ 1,186,800	\$ 1,187,400	\$ 1,164,240	\$ 1,101,700	\$ 10,948,340
Circulation	\$ 1,174,100	\$ 34,200									\$ 1,208,300
Cardiac	\$ 79,800	\$ 1,222,750	\$ 205,200	\$ 163,020	\$ 250,800	\$ 210,900	\$ 450,870	\$ 278,160	\$ 226,290	\$ 296,400	\$ 3,384,190
Cancer		\$ 181,260	\$ 1,219,050	\$ 139,650		\$ 181,830	\$ 27,930	\$ 136,230	\$ 153,330	\$ 186,960	\$ 2,226,240
Gas	\$ 98,040	\$ 146,490	\$ 99,750	\$ 1,190,550	\$ 320,340	\$ 188,100	\$ 202,350	\$ 216,600	\$ 396,720	\$ 267,900	\$ 3,126,840
Thyroid	\$ 93,480	\$ 649,800			\$ 1,225,200	\$ 190,380	\$ 829,350		\$ 555,180		\$ 3,543,390
Opiate	\$ 156,050	\$ 170,980	\$ 92,340	\$ 208,050	\$ 478,800	\$ 261,060	\$ 50,160	\$ 1,131,740	\$ 414,410	\$ 296,400	\$ 3,259,990
Motion		\$ 116,850	\$ 179,550	\$ 165,300	\$ 100,890	\$ 1,128,850	\$ 348,270	\$ 210,900	\$ 111,810	\$ 453,150	\$ 2,815,570
Proteins		\$ 434,910	\$ 105,450		\$ 741,000	\$ 506,730	\$ 1,237,600	\$ 119,700	\$ 320,340	\$ 477,090	\$ 3,942,820
Botulinum	\$ 180,750	\$ 520,980	\$ 71,500	\$ 248,520		\$ 102,600	\$ 701,100	\$ 397,290	\$ 973,510	\$ 228,000	\$ 3,424,250
Amphetamine			\$ 74,100	\$ 188,100		\$ 145,350				\$ 1,225,350	\$ 1,632,900
Total	\$ 2,947,420	\$ 4,645,620	\$ 2,879,440	\$ 3,081,290	\$ 4,305,630	\$ 4,092,200	\$ 5,034,430	\$ 3,678,020	\$ 4,315,830	\$ 4,532,950	\$ 39,512,830

Trading Game Results

Market Grid: Average Price and Penetration

	B612	Dune	Globex	Gotham	Metropolis	Panem	Serenity	Vulcan	Westeros	Zion	Average
Hormone	\$ 315	\$ 314	\$ 320	\$ 322	\$ 314	\$ 302	\$ 314	\$ 314	\$ 311	\$ 326	\$ 315
Circulation	\$ 482	\$ 570									\$ 484
Cardiac	\$ 570	\$ 502	\$ 570	\$ 570	\$ 570	\$ 570	\$ 570	\$ 570	\$ 570	\$ 570	\$ 544
Cancer		\$ 570	\$ 504	\$ 570		\$ 570	\$ 570	\$ 570	\$ 570	\$ 570	\$ 532
Gas	\$ 570	\$ 570	\$ 570	\$ 406	\$ 570	\$ 570	\$ 570	\$ 570	\$ 570	\$ 570	\$ 494
Thyroid	\$ 570	\$ 570			\$ 504	\$ 570	\$ 570		\$ 570		\$ 545
Opiate	\$ 527	\$ 566	\$ 570	\$ 570	\$ 570	\$ 570	\$ 570	\$ 513	\$ 550	\$ 570	\$ 544
Motion		\$ 570	\$ 570	\$ 570	\$ 570	\$ 490	\$ 570	\$ 570	\$ 568	\$ 570	\$ 535
Proteins		\$ 570	\$ 570		\$ 570	\$ 570	\$ 503	\$ 570	\$ 570	\$ 570	\$ 547
Botulinum	\$ 505	\$ 570	\$ 550	\$ 570		\$ 570	\$ 570	\$ 570	\$ 494	\$ 570	\$ 542
Amphetamine			\$ 570	\$ 570		\$ 570				\$ 475	\$ 496
Average	\$ 406	\$ 460	\$ 444	\$ 422	\$ 451	\$ 439	\$ 465	\$ 439	\$ 451	\$ 461	

	B612	Dune	Globex	Gotham	Metropolis	Panem	Serenity	Vulcan	Westeros	Zion	Total
Hormone	1.23%	1.24%	<1%	<1%	1.26%	1.30%	1.26%	1.26%	1.25%	1.13%	1.16%
Circulation	1.74%	<1%									0.21%
Cardiac	<1%	1.74%	<1%	<1%	<1%	<1%	<1%	<1%	<1%	<1%	0.59%
Cancer		<1%	1.73%	<1%		<1%	<1%	<1%	<1%	<1%	0.35%
Gas	<1%	<1%	<1%	2.09%	<1%	<1%	<1%	<1%	<1%	<1%	0.70%
Thyroid	<1%	<1%			1.74%	<1%	<1%		<1%		0.61%
Opiate	<1%	<1%	<1%	<1%	<1%	<1%	<1%	1.58%	<1%	<1%	0.58%
Motion		<1%	<1%	<1%	<1%	1.65%	<1%	<1%	<1%	<1%	0.41%
Proteins		<1%	<1%		<1%	<1%	1.76%	<1%	<1%	<1%	0.65%
Botulinum	<1%	<1%	<1%	<1%		<1%	<1%	<1%	1.41%	<1%	0.49%
Amphetamine			<1%	<1%		<1%				1.84%	0.31%
Total	0.49%	0.76%	0.45%	0.51%	0.67%	0.63%	0.81%	0.59%	0.67%	0.67%	

Income Statement

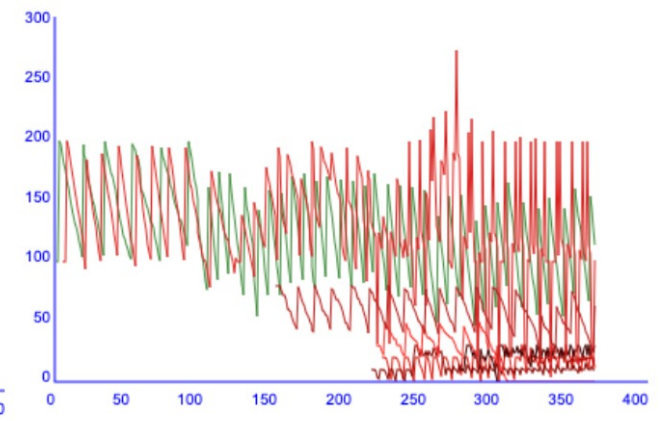
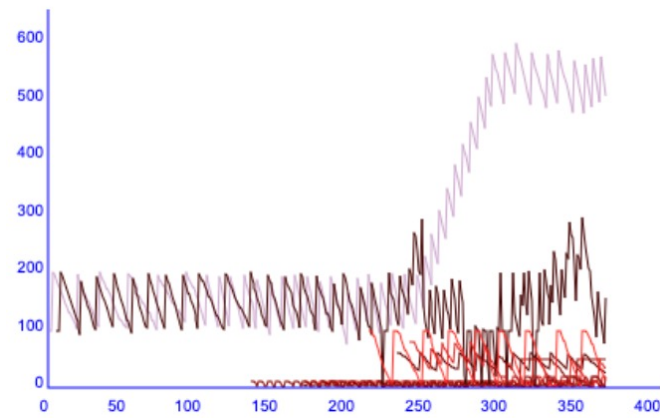
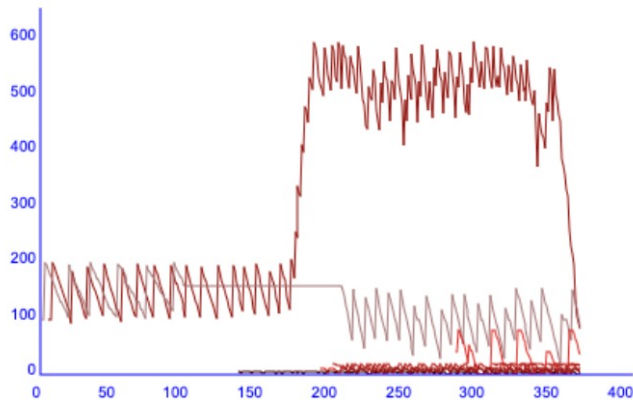
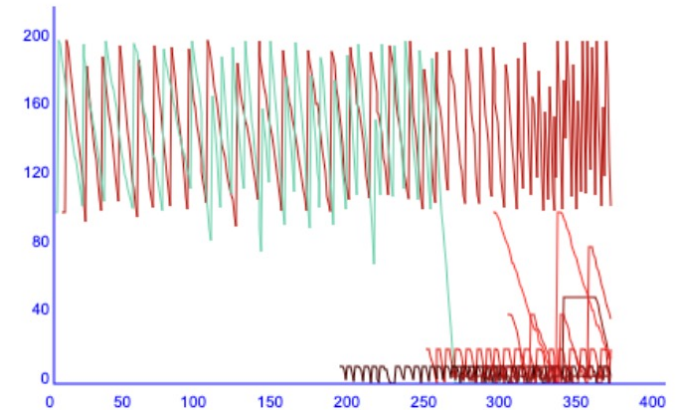
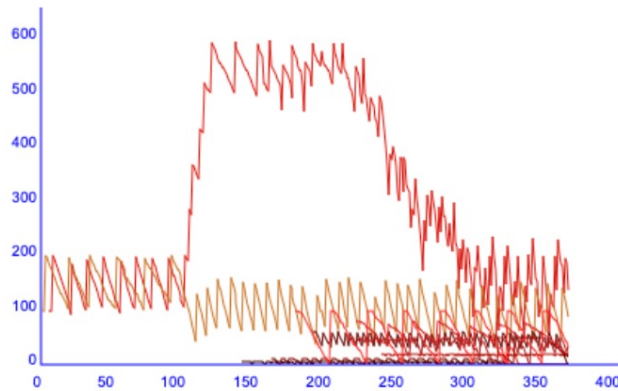
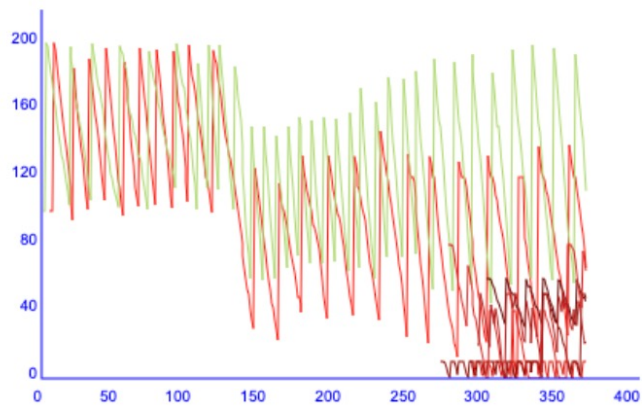
	\$	B612	Dune	Globex	Gotham	Metropolis	Panem	Serenity	Vulcan	Westeros	Zion
Sales revenue	\$	1,210,140	2,648,600	1,546,670	2,024,390	2,265,460	2,233,275	2,910,240	2,308,420	2,540,880	2,188,710
Cost of goods sold	\$	575,743	1,213,706	649,892	807,197	884,053	1,030,084	1,332,529	949,068	1,222,994	1,133,316
Gross profit	\$	634,397	1,434,894	896,778	1,217,193	1,381,407	1,203,191	1,577,711	1,359,352	1,317,886	1,055,394
Selling expense	\$	2,400	86,400	116,400	116,400	80,400	88,400	94,400	90,400	102,000	24,000
Total selling expense	\$	2,400	86,400	116,400	116,400	80,400	88,400	94,400	90,400	102,000	24,000
Distribution center depreciation expense	\$	41,667	41,667	41,667	41,667	41,667	41,667	41,667	41,667	41,667	41,667
Idle factory expense	\$	911	0	705	209	367	0	98	0	0	797
Other operating expenses	\$	42,578	41,667	42,372	41,876	42,033	41,667	41,765	41,667	41,667	42,463
Operating income	\$	589,419	1,306,828	738,007	1,058,917	1,258,974	1,073,124	1,441,546	1,227,285	1,174,219	988,931
Interest revenue	\$	10,345	16,389	11,494	13,293	18,029	14,782	18,574	17,437	16,118	13,384
Net interest and fees revenue (expense)	\$	10,345	16,389	11,494	13,293	18,029	14,782	18,574	17,437	16,118	13,384
Total net income before tax	\$	599,764	1,323,216	749,500	1,072,209	1,277,002	1,087,906	1,460,119	1,244,722	1,190,338	1,002,315
Income tax expense	\$	209,917	463,125	262,325	375,273	446,950	380,767	511,041	435,652	416,618	350,810
Net income after tax and before extraordinary items	\$	389,847	860,091	487,176	696,936	830,052	707,139	949,078	809,070	773,720	651,505
Net income	\$	389,847	860,091	487,176	696,936	830,052	707,139	949,078	809,070	773,720	651,505

Income Statement

	%	B612	Dune	Globex	Gotham	Metropolis	Panem	Serenity	Vulcan	Westeros	Zion
Sales revenue	%	100	100	100	100	100	100	100	100	100	100
Cost of goods sold	%	47.58	45.82	42.02	39.87	39.02	46.12	45.79	41.11	48.13	51.78
Gross profit	%	52.42	54.18	57.98	60.13	60.98	53.88	54.21	58.89	51.87	48.22
Selling expense	%	0.20	3.26	7.53	5.75	3.55	3.96	3.24	3.92	4.01	1.10
Total selling expense	%	0.20	3.26	7.53	5.75	3.55	3.96	3.24	3.92	4.01	1.10
Distribution center depreciation expense	%	3.44	1.57	2.69	2.06	1.84	1.87	1.43	1.80	1.64	1.90
Idle factory expense	%	0.08	0	0.05	0.01	0.02	0	0.00	0	0	0.04
Other operating expenses	%	3.52	1.57	2.74	2.07	1.86	1.87	1.44	1.80	1.64	1.94
Operating income	%	48.71	49.34	47.72	52.31	55.57	48.05	49.53	53.17	46.21	45.18
Interest revenue	%	0.85	0.62	0.74	0.66	0.80	0.66	0.64	0.76	0.63	0.61
Net interest and fees revenue (expense)	%	0.85	0.62	0.74	0.66	0.80	0.66	0.64	0.76	0.63	0.61
Total net income before tax	%	49.56	49.96	48.46	52.96	56.37	48.71	50.17	53.92	46.85	45.79
Income tax expense	%	17.35	17.49	16.96	18.54	19.73	17.05	17.56	18.87	16.40	16.03
Net income after tax and before extraordinary items	%	32.22	32.47	31.50	34.43	36.64	31.66	32.61	35.05	30.45	29.77
Net income	%	32.22	32.47	31.50	34.43	36.64	31.66	32.61	35.05	30.45	29.77

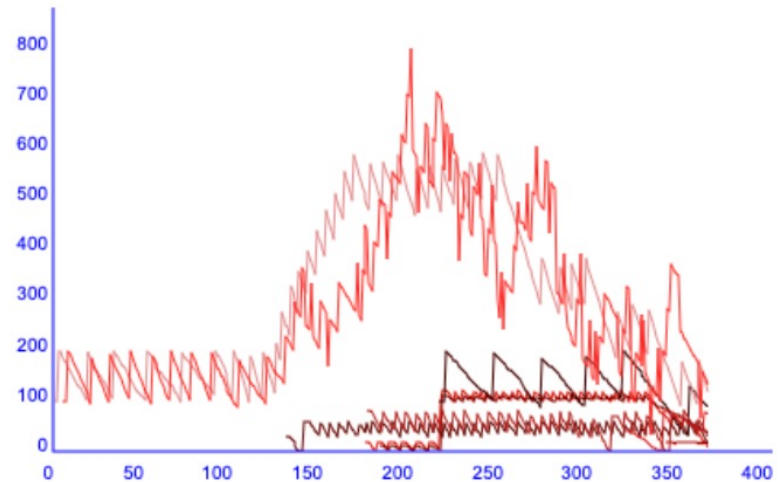
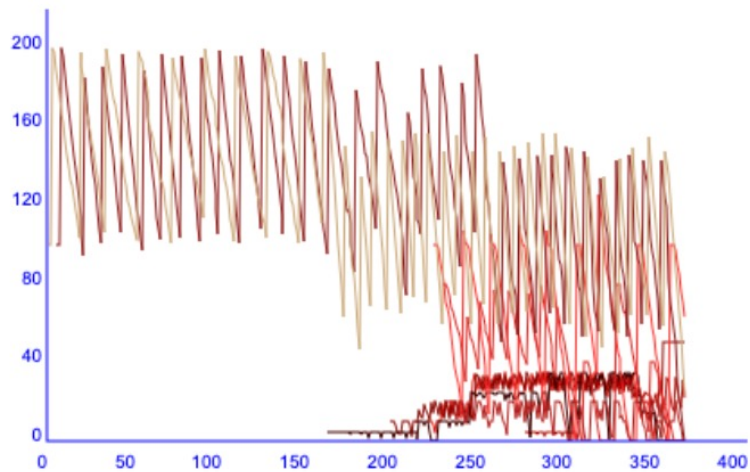
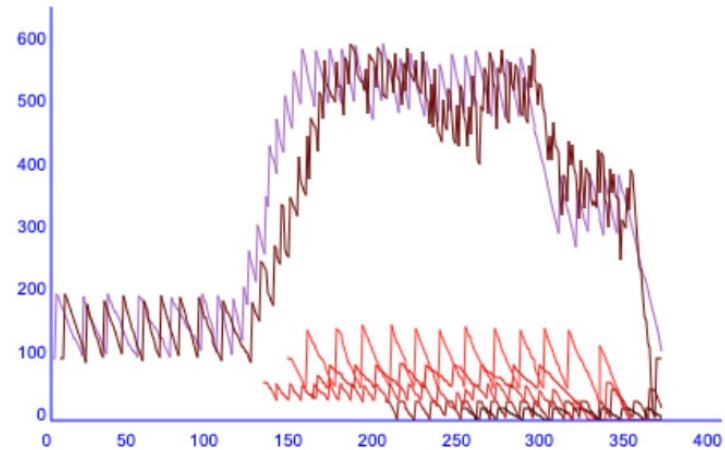
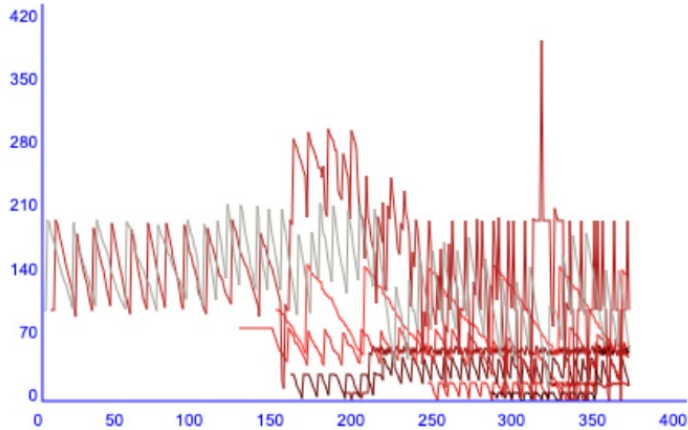
Trading Game Results

On Hand Inventory



Trading Game Results

On Hand Inventory



Working Capital

Consolidated Working Capital

Click on an account name to open a chart.

		B612	Dune	Globex	Gotham	Metropolis	Panem	Serenity	Vulcan	Westeros	Zion
Current assets:											
Cash	\$	1,467,302	2,399,097	1,756,041	1,962,973	2,644,542	2,169,325	2,728,983	2,543,009	2,326,490	1,958,344
Accounts receivable	\$	0	129,200	0	28,800	0	0	0	0	0	0
Work in process inventory	\$	0	10,040	0	0	0	10,010	0	10,017	10,028	0
Finished goods inventory	\$	89,921	57,707	46,863	42,617	28,586	116,524	129,519	61,293	153,315	53,171
Tax benefit	\$	0	0	0	0	0	0	0	0	0	0
Total current assets	\$	1,557,223	2,596,044	1,802,903	2,034,390	2,673,127	2,295,859	2,858,503	2,614,319	2,489,833	2,011,514
Current liabilities:											
Accounts payable	\$	0	0	0	0	0	0	0	158,000	0	0
Raw materials payable	\$	27,000	60,000	40,000	20,000	10,000	60,000	40,000	60,000	60,000	20,000
Taxes payable	\$	0	0	0	0	0	0	0	0	0	0
Other payables	\$	0	25,600	58,800	38,800	25,200	26,800	20,000	30,000	35,600	8,000
Total current liabilities	\$	27,000	85,600	98,800	58,800	35,200	86,800	60,000	248,000	95,600	28,000
Total working capital	\$	1,530,223	2,510,444	1,704,103	1,975,590	2,637,927	2,209,059	2,798,503	2,366,319	2,394,233	1,983,514
current ratio		57.67	30.33	18.25	34.60	75.94	26.45	47.64	10.54	26.04	71.84
Total non-cash working capital	\$	62,921	111,346	(51,937)	12,617	(6,614)	39,734	69,519	(176,690)	67,743	25,171
non-cash current ratio		3.33	2.30	0.47	1.21	0.81	1.46	2.16	0.29	1.71	1.90
quick ratio		54.34	29.54	17.77	33.87	75.13	24.99	45.48	10.25	24.34	69.94

Service Rates

Inventory Status Grid

Product features	Product										
	B612-Circulation	95.38%	73.55%								
	Dune-Cardiac	43.74%	98.64%	54.36%	50.38%	81.21%	45.29%	90.63%	64.80%	61.96%	85.16%
	Globex-Cancer		100%	98.61%	95.00%		32.43%	100%	59.16%	58.66%	71.52%
	Gotham-Gas	58.66%	58.56%	41.18%	100%	68.68%	44.25%	91.74%	60.05%	95.58%	87.74%
	Metropolis-Thyroid	100%	99.64%			99.05%	100%	98.27%		91.29%	
	Panem-Motion		26.39%	41.60%	16.59%	74.01%	91.08%	100.00%	89.28%	51.30%	43.32%
	Serenity-Proteins		88.67%	95.34%		100.00%	81.83%	99.56%	100%	100%	86.88%
	Vulcan-Opiate	64.72%	100%	76.51%	49.11%	92.09%	81.11%	89.69%	92.52%	97.92%	84.08%
	Westeros-Botulinum	86.19%	88.74%	18.11%	80.01%		48.40%	94.28%	100.00%	78.55%	54.76%
	Zion-Amphetamine			24.00%	38.97%		41.81%				99.39%
	B612-Hormone	98.78%									
	Dune-Hormone		98.81%								
	Globex-Hormone			100%							
	Gotham-Hormone				98.25%						
	Metropolis-Hormone					100%					
	Panem-Hormone						100%				
	Serenity-Hormone							100%			
	Vulcan-Hormone								99.99%		
	Westeros-Hormone									98.05%	
	Zion-Hormone										99.91%

Production Game

Moving on!

Production Game

We are not monopolists! It is not a one-shot game!

- **Do not use monopoly pricing – that was only for the production and trading games.**
- **Daily production is no longer given at 40/day**
- Leadtime is no longer 3.5 days
- Production cost is no longer \$100
- You will have to pay for shipping
- You will have to pay for overhead
- Retailers will have to pay handling fees
- Retailers will have to pay 20% commissions to DC salespeople.
- Put yesterday behind you – we are no longer monopolists.

Set-up

- Bench factory
 - \$100,000 land, \$100,000 capital investment, \$2,500/day in recurring labor costs
- Heart View product
- Design new goods
 - Gleacher Game Market Research for Production Game (PDF) and Product Design Guide (Excel)
 - R&D days and cost, materials cost
 - Consider willingness to pay
- Manage your factory
 - Recurring costs
 - Batch sizes
 - Priorities
 - Production function
- **We will not build new factories until Tuesday**
- You will have plenty of cash to design products today

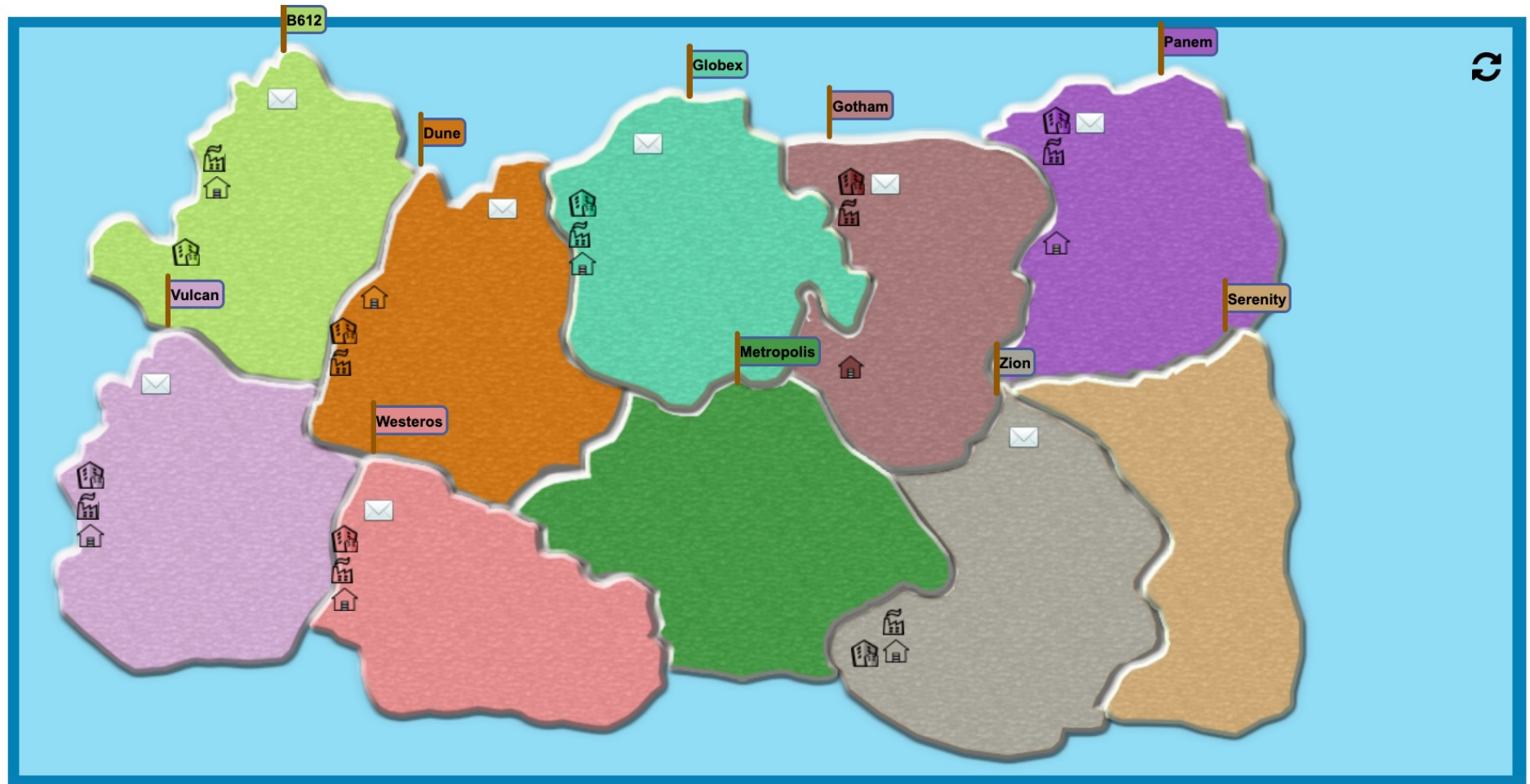
Production Game

Play

- Start at the end of quarter 4 (day 365).
- Play 1 hour in class to the end of quarter 6 (day 546).
- Play 2 hours this evening to the end of quarter 10 (day 910).
- We'll play the game without you to day 1092, to see what would have happened.
- Speed: 3 game days per minute.
- You may want to work on homework concurrently, so that you connect the homework to the activities in the game. That's not a requirement, just a suggestion. At a minimum, please review the assignment questions before you play tonight.
- Catherine and Kathleen will not be playing tonight so they cannot trade with you.

Production Game Map

Please pay attention to envelopes!



Starting Point

Goal now: Maximize Current Shareholder Value

World	Team Name	Return to Investors	Retained Earnings at end of day 364	Cash Position
1	B612	\$ 1,579,530	\$ -342,398	\$ 1,579,530
1	Dune	\$ 1,579,530	\$ -342,398	\$ 1,579,530
1	Globex	\$ 1,579,530	\$ -342,398	\$ 1,579,530
1	Gotham	\$ 1,579,530	\$ -342,398	\$ 1,579,530
1	Panem	\$ 1,579,530	\$ -342,398	\$ 1,579,530
1	Vulcan	\$ 1,579,530	\$ -342,398	\$ 1,579,530
1	Westeros	\$ 1,579,530	\$ -342,398	\$ 1,579,530
1	Zion	\$ 1,579,530	\$ -342,398	\$ 1,579,530

Cash Balance:	1,579,530
- Unsecured Debt:	0
+ Paid Dividends:	0
+ Subsequent Returns:	0
= Return to Investors:	1,579,530

Starting Point

Income Statement

	\$	B612	Dune	Globex	Gotham	Panem	Vulcan	Westeros	Zion
Sales revenue	\$	682,500	682,500	682,500	682,500	682,500	682,500	682,500	682,500
Cost of goods sold	\$	417,743	417,743	417,743	417,743	417,743	417,743	417,743	417,743
Gross profit	\$	264,757	264,757	264,757	264,757	264,757	264,757	264,757	264,757
Selling expense	\$	146,250	146,250	146,250	146,250	146,250	146,250	146,250	146,250
Distribution center operating expense	\$	182,000	182,000	182,000	182,000	182,000	182,000	182,000	182,000
Total selling expense	\$	328,250	328,250	328,250	328,250	328,250	328,250	328,250	328,250
Distribution center depreciation expense	\$	41,667	41,667	41,667	41,667	41,667	41,667	41,667	41,667
Idle factory expense	\$	0	0	0	0	0	0	0	0
Other operating expenses	\$	41,667	41,667	41,667	41,667	41,667	41,667	41,667	41,667
Operating income	\$	(105,160)	(105,160)	(105,160)	(105,160)	(105,160)	(105,160)	(105,160)	(105,160)
Interest revenue	\$	11,913	11,913	11,913	11,913	11,913	11,913	11,913	11,913
Net interest and fees revenue (expense)	\$	11,913	11,913	11,913	11,913	11,913	11,913	11,913	11,913
Total net income before tax	\$	(93,247)	(93,247)	(93,247)	(93,247)	(93,247)	(93,247)	(93,247)	(93,247)
Income tax expense	\$	(32,636)	(32,636)	(32,636)	(32,636)	(32,636)	(32,636)	(32,636)	(32,636)
Net income after tax and before extraordinary items	\$	(60,611)	(60,611)	(60,611)	(60,611)	(60,611)	(60,611)	(60,611)	(60,611)
Net income	\$	(60,611)	(60,611)	(60,611)	(60,611)	(60,611)	(60,611)	(60,611)	(60,611)

Starting Point

Balance Sheet

	\$	B612	Dune	Globex	Gotham	Panem	Vulcan	Westeros	Zion
Assets									
Current assets:									
Cash	\$	1,579,530	1,579,530	1,579,530	1,579,530	1,579,530	1,579,530	1,579,530	1,579,530
Accounts receivable	\$	240,100	240,100	240,100	240,100	240,100	240,100	240,100	240,100
Work in process inventory	\$	33,157	33,157	33,157	33,157	33,157	33,157	33,157	33,157
Finished goods inventory	\$	112,683	112,683	112,683	112,683	112,683	112,683	112,683	112,683
Tax benefit	\$	184,365	184,365	184,365	184,365	184,365	184,365	184,365	184,365
Total current assets	\$	2,149,836	2,149,836	2,149,836	2,149,836	2,149,836	2,149,836	2,149,836	2,149,836
Noncurrent assets:									
Land	\$	200,000	200,000	200,000	200,000	200,000	200,000	200,000	200,000
Property plant and equipment - factories:									
Property plant and equipment - factories	\$	100,000	100,000	100,000	100,000	100,000	100,000	100,000	100,000
Accumulated depreciation - factories	\$	(6,667)	(6,667)	(6,667)	(6,667)	(6,667)	(6,667)	(6,667)	(6,667)
Net Property plant and equipment - factories	\$	93,333	93,333	93,333	93,333	93,333	93,333	93,333	93,333
Property plant and equipment - distribution centers:									
Property plant and equipment - distribution centers	\$	2,500,000	2,500,000	2,500,000	2,500,000	2,500,000	2,500,000	2,500,000	2,500,000
Accumulated depreciation - distribution centers	\$	(166,667)	(166,667)	(166,667)	(166,667)	(166,667)	(166,667)	(166,667)	(166,667)
Net Property plant and equipment - distribution centers	\$	2,333,333	2,333,333	2,333,333	2,333,333	2,333,333	2,333,333	2,333,333	2,333,333
Total assets	\$	4,776,502	4,776,502	4,776,502	4,776,502	4,776,502	4,776,502	4,776,502	4,776,502
Liabilities and shareholders' equity									
Liabilities:									
Current liabilities:									
Raw materials payable	\$	20,000	20,000	20,000	20,000	20,000	20,000	20,000	20,000
Other payables	\$	98,900	98,900	98,900	98,900	98,900	98,900	98,900	98,900
Total current liabilities	\$	118,900	118,900	118,900	118,900	118,900	118,900	118,900	118,900
Noncurrent liabilities:									
Noncurrent liabilities	\$	0	0	0	0	0	0	0	0
Shareholder equity:									
Contributed capital	\$	5,000,000	5,000,000	5,000,000	5,000,000	5,000,000	5,000,000	5,000,000	5,000,000
Retained earnings	\$	(342,398)	(342,398)	(342,398)	(342,398)	(342,398)	(342,398)	(342,398)	(342,398)
Total shareholder equity	\$	4,657,602	4,657,602	4,657,602	4,657,602	4,657,602	4,657,602	4,657,602	4,657,602
Total liabilities and shareholders' equity	\$	4,776,502	4,776,502	4,776,502	4,776,502	4,776,502	4,776,502	4,776,502	4,776,502

Starting Point

Working Capital

		B612	Dune	Globex	Gotham	Panem	Vulcan	Westeros	Zion
Current assets:									
Cash	\$	1,579,530	1,579,530	1,579,530	1,579,530	1,579,530	1,579,530	1,579,530	1,579,530
Accounts receivable	\$	240,100	240,100	240,100	240,100	240,100	240,100	240,100	240,100
Work in process inventory	\$	33,157	33,157	33,157	33,157	33,157	33,157	33,157	33,157
Finished goods inventory	\$	112,683	112,683	112,683	112,683	112,683	112,683	112,683	112,683
Tax benefit	\$	184,365	184,365	184,365	184,365	184,365	184,365	184,365	184,365
Total current assets	\$	2,149,836	2,149,836	2,149,836	2,149,836	2,149,836	2,149,836	2,149,836	2,149,836
Current liabilities:									
Raw materials payable	\$	20,000	20,000	20,000	20,000	20,000	20,000	20,000	20,000
Other payables	\$	98,900	98,900	98,900	98,900	98,900	98,900	98,900	98,900
Total current liabilities	\$	118,900	118,900	118,900	118,900	118,900	118,900	118,900	118,900
Total working capital	\$	2,030,936	2,030,936	2,030,936	2,030,936	2,030,936	2,030,936	2,030,936	2,030,936
current ratio		18.08	18.08	18.08	18.08	18.08	18.08	18.08	18.08
Total non-cash working capital	\$	451,406	451,406	451,406	451,406	451,406	451,406	451,406	451,406
non-cash current ratio		4.80	4.80	4.80	4.80	4.80	4.80	4.80	4.80
quick ratio		15.30	15.30	15.30	15.30	15.30	15.30	15.30	15.30

Headquarters

B612 Headquarters



Hover over values for details. Click on numbers and icons to open panels and edit values.

As of day 364 ↺

0 days to Q4

Finance

Operating Income : \$-105,160 (Q 4)

Cash : \$1,579,530

Working capital ratio : 18.08

Trade credit : materials net / 30
others net / 15

Dividends : \$0

Marketing

Designs : 0 ... 0 1 0

Retail sales : 1 of 50 regional markets
1 inventories
\$0 per day advertising
2,837 units sold total
\$1,985,900 total revenues

Sales to other teams : 0 inventories
0 units sold total
\$0 total revenues

Operations

Facilities : +

All agreements : 0 ... 1 0
263 units on hand
0 units in transit

Production : 0 ... 1 0
100 units in process

Procurement : 0 ... 0 0

Fulfillment : 0 ... 0 0

Factory

Pilot Factory in Dune [Dune] [PROCESSING]



Name : Pilot Factory in Dune

Owner : Dune

Location : Dune

Status : PROCESSING

Technology : Bench

Land expenditure : \$ 100,000

Capital expenditure : \$ 100,000

Daily expenditure rate : \$ 2,500

Operating income : \$ -1,071,667

Inventory in process : 100 Units
of Dune-Heart View
for Local DC in Dune

Inventory in transit : 0 Units
of 0 Products
for 0 DCs

Throughput : 9.04 units per day

Overhead per Unit : \$ 278.45

Suspended on day 364

1 Outgoing Shipping Agreement



ADD NEW +

10 records

Search:

Id	Destination	Product	Priority	Terms	Days delayed	Batches	Projected contribution per unit	Status
		Name					Margin	
2		Heart View	0	100 @ \$100	0	32	\$ -50	ACTIVE

Showing 1 to 1 of 1 entries

< 1 >

Increase Capital in your Factory

Pilot Factory in Globex [Globex] [PROCESSING]

Name : Pilot Factory in Globex

Owner : Globex

Location : Globex

Status : PROCESSING

Technology : Bench

Land expenditure : \$ 100,000

Daily expenditure : \$ 2,500

Capital expenditure : \$ 100,000

Operating income : \$ -1,081,703

Inventory in process : 100 Units

Inventory in transit : 100 Units

Throughput : 9.04 units per day

Overhead per Unit : \$ 278.45

As of day 366.14

1 Outgoing Shipping Agreement

10 records

Search:

Id	Destination	Product	Priority	Terms	Days delayed	Batches	Projected contribution per unit	Status
		Name					Margin	
1		Heart View	0	100 @ \$100	0	33	\$ -50	ACTIVE

Showing 1 to 1 of 1 entries

Capital expenditure : \$ 100,000

\$

✓

✗

Confirm Transaction

Pending Capital Expenditure will be effective on day 411.00

Factory Name : Pilot Factory in Arryn

Current Cap Exp : 100,000

Requested Cap Exp : 1,000.00

Effective Day : 411.00

When : Now

Confirm

Cancel

CHICAGO BOOTH

Integrated Strategic Management
Creating Value in Complex and Ambiguous Settings

30

Change Labor (Daily Expenditures) in your Factory

Pilot Factory in Globex [Globex] [PROCESSING]

Name : Pilot Factory in Globex
Owner : Globex
Location : Globex
Status : PROCESSING
Technology : Bench
Land expenditure : \$ 100,000
Daily expenditure : \$ 2,500
Capital expenditure : \$ 100,000

Operating income : \$ -1,081,703
Inventory in process : 100 Units
of Globex-Heart View
for Local DC in Globex
Inventory in transit : 100 Units
of Globex-Heart View
for Local DC in Globex
Throughput : 9.04 units per day
Overhead per Unit : \$ 278.45

1 Outgoing Shipping Agreement

10 records

Search:

Id	Destination	Product	Priority	Terms	Days delayed	Batches	Projected contribution per unit	Status
		Name					Margin	
1		Heart View	0	100 @ \$100	0	33	\$ -50	ACTIVE

Showing 1 to 1 of 1 entries

Close

Daily expenditure : \$ 2,500

\$ 0

✓ ✗

Confirm Transaction

Factory Change Daily Exp Transaction
Factory Name : Pilot Factory in Arrakis
Current Daily Exp : 2,500.00
Requested Daily Exp : 2,000.00
When : Now

Confirm

Cancel

Distribution Center

Local DC in Serenity [Serenity] [READY]



Name : Local DC in Serenity
Owner : Serenity
Location : Serenity
Status : READY
Operating income : \$ 381,983

Total advertising : \$ 0
Total inventory : 363 units
On-hand inventory : 263 units
Price : \$ 700
Total retail sales : 2,837 units

Suspended on day 364

1 Inventory

10 records

Include Inactive: ☐ Search:

Product	\$	Units	Advertising	Last date															
Name	Price	On hand	Daily \$	Any															
Heart View	700	263	0	363.91															

Showing 1 to 1 of 1 entries

1 Incoming Shipping Agreement

10 records

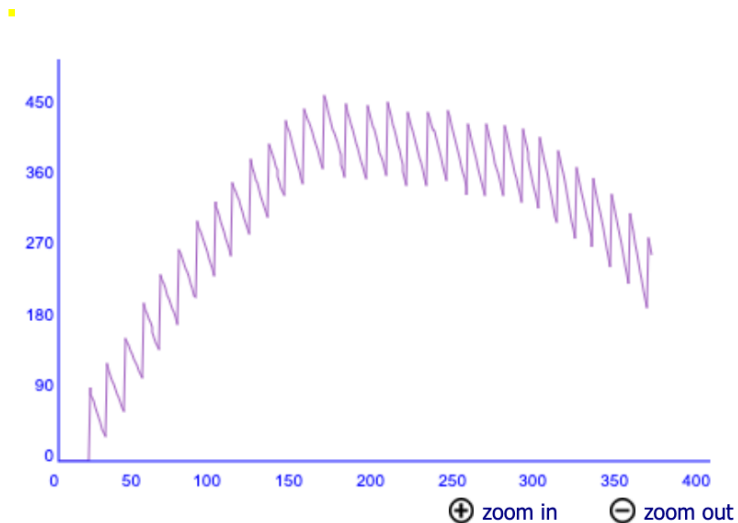
Search:

Id	Origin	Product	Reorder point	Terms	Days delayed	Batches	Projected contribution per unit	Status	
		Name					Margin		
6		Heart View	500	100 @ \$100	0	32	\$ 450	ACTIVE	

Showing 1 to 1 of 1 entries

Inventory and Sales

End-of-day On-hand Inventory at Local DC in Metropolis

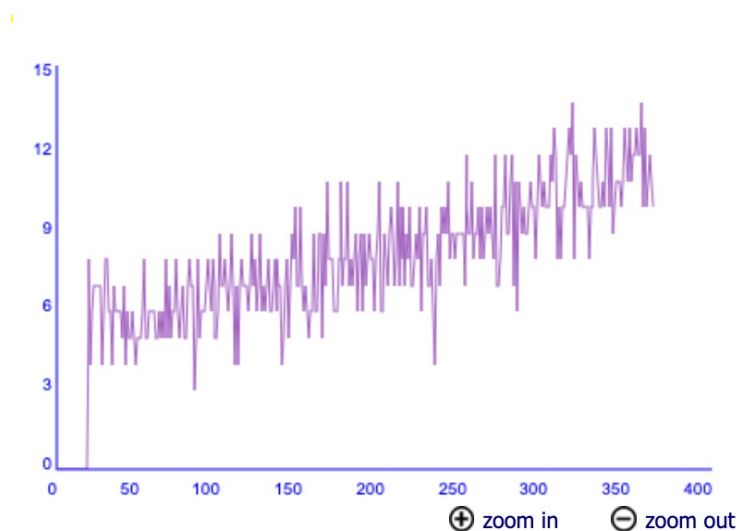


drag mouse over region of interest to zoom in plot

Reset

Data

Daily Retail Sales from Local DC in Metropolis



drag mouse over region of interest to zoom in plot

Reset

Data

Market Grid

Only 5 markets!

	B612	Dune	Globex	Gotham	Metropolis	Panem	Serenity	Vulcan	Westeros	Zion	Total
MD - Heart	2,837	2,837	2,837	2,837		2,837		2,837	2,837	2,837	22,696
MD - Breast											
MD - Estrogen											
Law - Narcotic											
Clinic - Fertility											
Total	2,837	2,837	2,837	2,837		2,837		2,837	2,837	2,837	22,696

Reports and General Information

From today on, we play the Gleacher Game, so please start learning it!

- Understand what is in the different reports and charts
- Go to your HQ and click on buttons to see what is there
- Read the Gleacher Game Player Manual
- Read the Gleacher Game Case Brief

Integrated Strategic Management

Creating Value in Complex
and Ambiguous Settings

Factories (production)

Manage Your Factory

Production Model: Cobb Douglas

Questions

- How many units can our factory produce in 1 day?
- How many units are in a batch of inventory?
- How long does it take to produce a batch of inventory?
- What does it cost us to produce a batch of inventory?
- What is the manufacturing overhead cost for each unit produced?
- What determines the answers to these questions?

Steps

- Use the Cobb Douglas production function to solve for daily production (throughput)
- Use Little's Law to get cycle time (CT)
- Add set up time to get CT with set up time
- Use Little's Law to get daily throughput adjusted for set up time

Cobb Douglas Production Function

$$Y = AK^{\alpha}L^{\beta}$$

Where:

- Y is the quantity produced over the year
- A is total factory productivity
- K is the units of capital employed (the capital invested in your factory)
- L is the daily recurring costs * 364
- α is the coefficient on capital
- β is the coefficient on labor

Current Technology and Inputs:

- Batch size = 100 units
- Daily recurring costs (labor) = \$2,500
- Factory cost (capital) = \$100,000
- $A = 0.009$
- $\alpha = 0.10$
- $\beta = 0.85$
- Setup time = 0.05 days

What is our daily throughput?

Little's Law

Little's law states that the average number of items within a system (inventory) is equal to the average arrival rate of items into and out of the system (throughput) multiplied by the average amount of time an item spends in the system:

$$INV = \lambda * CT$$

INV = inventory, number of jobs in the system

λ = throughput, rate at which jobs pass through the system

CT = cycle time, the time a job remains in a system

Notes and Calculations

Notes and Calculations

All Factories

Technology	A	α	β	Technology Description
Benches	0.009	0.10	0.85	Lab benches with general-purpose equipment. Each table is manned by a skilled worker who produces an entire product. Setup time for new batches = 0.05 days
Production Line	0.010	0.30	0.75	A row of tables with specialized equipment and each worker assigned a specific portion of the production sequence. Workers are relatively unskilled. Setup time for new batches = 0.50 days
Automated Cell	0.020	0.80	0.30	A robot that feeds material to automated production machines clustered around it. There is support staff but no workers performing actual production. Setup time for new batches = 1 day

Production Model Template

Bench Factory Parameters		$Y = A K^{\alpha} L^{\beta}$	
A	0.0090	Depreciation years	15
α	0.10	# of days per year	364
β	0.85	Materials cost per unit	
$\alpha + \beta$	0.95		
Setup time, in days	0.05		
Batch size			
Minimum capital	\$ 1		

Daily Throughput in units, including setup time								
K ↓ vs L _{daily} →	500	1,000	2,500	3,000	4,000	8,000	16,000	
100,000								
200,000								
300,000								
400,000								
500,000								
600,000								
700,000								
800,000								
900,000								
1,000,000								
2,000,000								
3,000,000								
4,000,000								
5,000,000								

Manufacturing Overhead per unit								
K ↓ vs L _{daily} →	500	1,000	2,500	3,000	4,000	8,000	16,000	
100,000								
200,000								
300,000								
400,000								
500,000								
600,000								
700,000								
800,000								
900,000								
1,000,000								
2,000,000								
3,000,000								
4,000,000								
5,000,000								

Production Model

Bench Factory Filled In – Use for check figures for tonight's assignment

Bench Factory Parameters	
A	0.0090
α	0.10
β	0.85
$\alpha+\beta$	0.95
Setup time, in days	0.05
Batch size	100
Minimum capital	\$ 1

$$Y = A K^{\alpha} L^{\beta}$$

Depreciation years	15
# of days per year	364
Materials cost per unit	\$ 100

Daily Throughput in units, including setup time							
K ↓ vs L _{daily} →	500	1,000	2,500	3,000	4,000	8,000	16,000
100,000	2.310	4.161	9.044	10.552	13.455	24.123	43.065
200,000	2.476	4.459	9.690	11.305	14.414	25.832	46.085
300,000	2.578	4.643	10.089	11.770	15.006	26.887	47.946
400,000	2.654	4.778	10.382	12.112	15.441	27.661	49.311
500,000	2.713	4.886	10.615	12.383	15.786	28.276	50.396
600,000	2.763	4.975	10.809	12.610	16.075	28.789	51.299
700,000	2.806	5.052	10.976	12.804	16.322	29.229	52.075
800,000	2.844	5.120	11.123	12.975	16.540	29.616	52.757
900,000	2.877	5.181	11.254	13.128	16.734	29.962	53.365
1,000,000	2.908	5.235	11.372	13.266	16.910	30.275	53.915
2,000,000	3.116	5.610	12.184	14.211	18.113	32.412	57.673
3,000,000	3.245	5.841	12.685	14.795	18.855	33.731	59.988
4,000,000	3.340	6.011	13.052	15.224	19.400	34.698	61.685
5,000,000	3.415	6.147	13.345	15.564	19.833	35.467	63.033

Manufacturing Overhead per unit							
K ↓ vs L _{daily} →	500	1,000	2,500	3,000	4,000	8,000	16,000
100,000	224.333	244.742	278.453	286.043	298.640	332.391	371.956
200,000	216.725	232.495	261.780	268.609	280.046	311.109	347.981
300,000	215.227	227.222	253.244	259.552	270.219	299.588	334.854
400,000	216.033	224.626	247.863	253.747	263.799	291.869	325.957
500,000	218.022	223.430	244.147	249.660	259.183	286.165	319.305
600,000	220.717	223.084	241.454	246.630	255.677	281.704	314.039
700,000	223.872	223.306	239.449	244.312	252.919	278.083	309.710
800,000	227.347	223.928	237.937	242.503	250.699	275.067	306.056
900,000	231.054	224.848	236.794	241.077	248.882	272.505	302.910
1,000,000	234.934	225.996	235.937	239.948	247.379	270.297	300.159
2,000,000	277.998	243.549	235.259	236.873	241.065	258.121	283.776
3,000,000	323.410	265.252	240.407	239.907	241.286	253.460	275.878
4,000,000	369.097	288.219	247.664	245.185	243.949	251.672	271.259
5,000,000	414.601	311.675	255.959	251.583	247.852	251.378	268.363

Integrated Strategic Management

Creating Value in Complex
and Ambiguous Settings

Accounting

Financial and Managerial Accounting

Accounting

Main Financial Statements

- Balance Sheet
- Income Statement
- Statement of Cash Flows

Financial Statement Analysis

- Working Capital

Managerial Accounting

- Contribution Margin
- Break-even Point

Balance Sheet

Gleacher Game

	\$	Quarter 4	Quarter 3	Quarter 2	Quarter 1	Quarter 0
Assets						
Current assets:						
Cash	\$	1,579,530	1,613,017	1,723,448	1,909,244	2,200,000
Accounts receivable	\$	240,100	191,800	158,900	126,700	0
Work in process inventory	\$	33,157	26,754	36,283	16,403	10,000
Finished goods inventory	\$	112,683	187,663	177,380	156,814	0
Tax benefit	\$	184,365	151,729	105,246	54,415	0
Total current assets	\$	2,149,836	2,170,963	2,201,257	2,263,576	2,210,000
Noncurrent assets:						
Land	\$	200,000	200,000	200,000	200,000	200,000
Property plant and equipment - factories:						
Property plant and equipment - factories	\$	100,000	100,000	100,000	100,000	100,000
Accumulated depreciation - factories	\$	(6,667)	(5,000)	(3,333)	(1,667)	0
Net Property plant and equipment - factories	\$	93,333	95,000	96,667	98,333	100,000
Property plant and equipment - distribution centers:						
Property plant and equipment - distribution centers	\$	2,500,000	2,500,000	2,500,000	2,500,000	2,500,000
Accumulated depreciation - distribution centers	\$	(166,667)	(125,000)	(83,333)	(41,667)	0
Net Property plant and equipment - distribution centers	\$	2,333,333	2,375,000	2,416,667	2,458,333	2,500,000
Total assets	\$	4,776,502	4,840,963	4,914,590	5,020,243	5,010,000
Liabilities and shareholders' equity						
Liabilities:						
Current liabilities:						
Raw materials payable	\$	20,000	30,000	20,000	30,000	10,000
Other payables	\$	98,900	92,750	90,050	91,300	0
Total current liabilities	\$	118,900	122,750	110,050	121,300	10,000
Noncurrent liabilities:						
Noncurrent liabilities	\$	0	0	0	0	0
Shareholder equity:						
Contributed capital	\$	5,000,000	5,000,000	5,000,000	5,000,000	5,000,000
Retained earnings	\$	(342,398)	(281,787)	(195,460)	(101,057)	0
Total shareholder equity	\$	4,657,602	4,718,213	4,804,540	4,898,943	5,000,000
Total liabilities and shareholders' equity	\$	4,776,502	4,840,963	4,914,590	5,020,243	5,010,000

Income Statement

Gleacher Game

	\$	Quarter 4	Quarter 3	Quarter 2	Quarter 1
Sales revenue	\$	682,500	543,200	456,400	303,800
Cost of goods sold	\$	417,743	332,481	279,352	185,949
Gross profit	\$	264,757	210,719	177,048	117,851
Selling expense	\$	146,250	116,400	97,800	65,100
Distribution center operating expense	\$	182,000	182,000	182,000	182,000
Total selling expense	\$	328,250	298,400	279,800	247,100
Distribution center depreciation expense	\$	41,667	41,667	41,667	41,667
Idle factory expense	\$	0	15,932	14,369	0
Other operating expenses	\$	41,667	57,598	56,036	41,667
Operating income	\$	(105,160)	(145,279)	(158,788)	(170,916)
Interest revenue	\$	11,913	12,469	13,554	15,444
Net interest and fees revenue (expense)	\$	11,913	12,469	13,554	15,444
Total net income before tax	\$	(93,247)	(132,810)	(145,234)	(155,472)
Income tax expense	\$	(32,636)	(46,483)	(50,831)	(54,415)
Net income after tax and before extraordinary items	\$	(60,611)	(86,327)	(94,402)	(101,057)
Net income	\$	(60,611)	(86,327)	(94,402)	(101,057)

Statement of Cash Flows

Gleacher Game

	Quarter 4	Quarter 3	Quarter 2	Quarter 1	Quarter 0
Cash flows from operating activities:					
Net Income	\$ (60,611)	(86,327)	(94,402)	(101,057)	0
Depreciation Expense	\$ 43,333	43,333	43,333	43,333	0
Adjustments to reconcile net income to net cash provided from operating activities:					
(Increase) Decrease in Accounts receivable	\$ (48,300)	(32,900)	(32,200)	(126,700)	0
(Increase) Decrease in Work in process inventory	\$ (6,403)	9,528	(19,879)	(6,403)	(10,000)
(Increase) Decrease in Finished goods inventory	\$ 74,980	(10,283)	(20,566)	(156,814)	0
(Increase) Decrease in Tax benefit	\$ (32,636)	(46,483)	(50,831)	(54,415)	0
(Decrease) Increase in Raw materials payable	\$ (10,000)	10,000	(10,000)	20,000	10,000
(Decrease) Increase in Other payables	\$ 6,150	2,700	(1,250)	91,300	0
Net cash from operating activities	\$ (33,487)	(110,431)	(185,796)	(290,756)	0
Cash flows from investing activities:					
Construct Factory Pilot Factory in Globex	\$ 0	0	0	0	(200,000)
Construct Distribution Center Local DC in Globex	\$ 0	0	0	0	(2,600,000)
Net cash from investing activities	\$ 0	0	0	0	(2,800,000)
Cash flows from financing activities:					
Issue stock	\$ 0	0	0	0	5,000,000
Net cash from financing activities	\$ 0	0	0	0	5,000,000
Net increase (decrease) in cash	\$ (33,487)	(110,431)	(185,796)	(290,756)	2,200,000
Cash, beginning of period	\$ 1,613,017	1,723,448	1,909,244	2,200,000	0
Cash, end of period	\$ 1,579,530	1,613,017	1,723,448	1,909,244	2,200,000

Working Capital

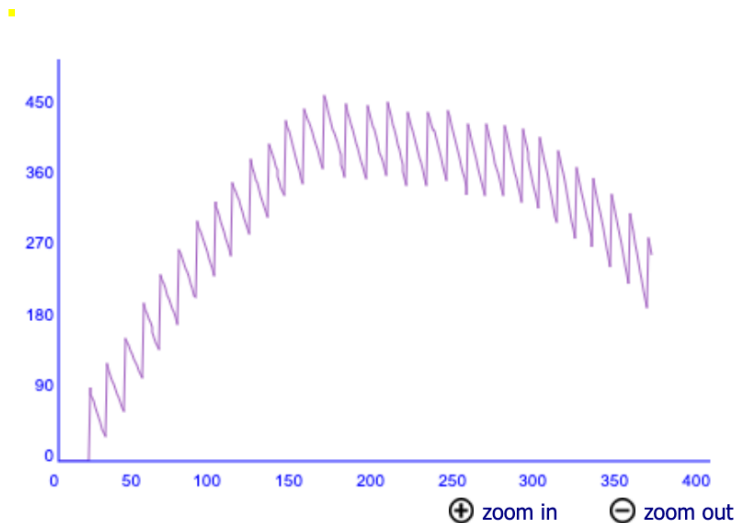
Gleacher Game

		Quarter 4	Quarter 3	Quarter 2	Quarter 1	Quarter 0
Current assets:						
Cash	\$	1,579,530	1,613,017	1,723,448	1,909,244	2,200,000
Accounts receivable	\$	240,100	191,800	158,900	126,700	0
Work in process inventory	\$	33,157	26,754	36,283	16,403	10,000
Finished goods inventory	\$	112,683	187,663	177,380	156,814	0
Tax benefit	\$	184,365	151,729	105,246	54,415	0
Total current assets	\$	2,149,836	2,170,963	2,201,257	2,263,576	2,210,000
Current liabilities:						
Raw materials payable	\$	20,000	30,000	20,000	30,000	10,000
Other payables	\$	98,900	92,750	90,050	91,300	0
Total current liabilities	\$	118,900	122,750	110,050	121,300	10,000
Total working capital	\$	2,030,936	2,048,213	2,091,207	2,142,276	2,200,000
Current ratio		18.08	17.69	20.00	18.66	221
Total Non-cash working capital	\$	451,406	435,196	367,759	233,032	0
Non-cash current ratio		4.80	4.55	4.34	2.92	1
Quick ratio		15.30	14.70	17.10	16.78	220

Inventory and Sales

Remember this when looking at financials

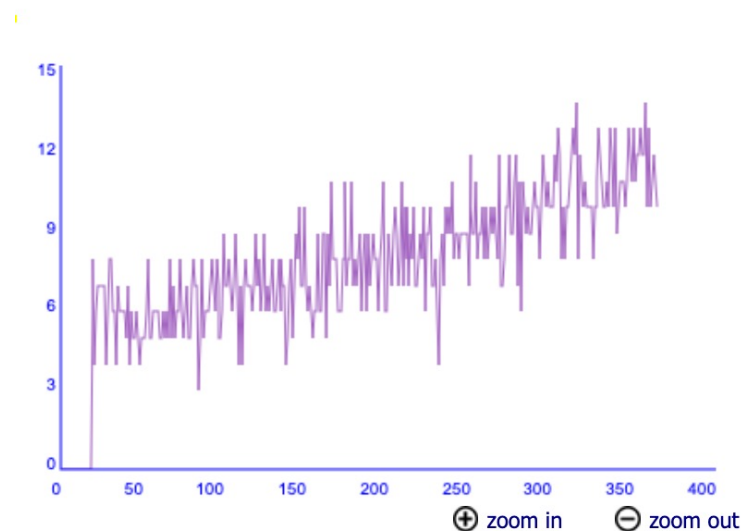
End-of-day On-hand Inventory at Local DC in Metropolis



drag mouse over region of interest to zoom in plot

Reset Data

Daily Retail Sales from Local DC in Metropolis



drag mouse over region of interest to zoom in plot

Reset Data

Contribution Margin for Heart View

Example

Contribution Margin = Price – Variable Costs = P – VC

Price = \$700

Variable costs:

- Direct materials (\$100)
- Manufacturing overhead (use what we calculated earlier)
- Shipping (to DC)
- Sales commission (20% of sales price)
- Handling (\$10 per unit)
- Taxes (35%)

Contribution Margin for Heart View

Contribution Margin = Price – Variable Cost = P – VC

Variable Cost = Manufacturing Overhead + Materials + Shipping to DC + Handling + Commission

	Per Unit
Revenue	
Manufacturing Overhead	
Materials	
Shipping	
Handling (\$10 per unit)	
Commission (20% of revenue)	
Contribution Margin before Tax	
Tax (35% tax rate)	
Contribution Margin after Tax	

Break-even Point

Heart View

Break-even point = Fixed Costs/Contribution Margin

Fixed costs = Distribution Center Operating Expense + Distribution Center Depreciation

	\$ Quarter 4
Sales revenue	\$ 682,500
Cost of goods sold	\$ 417,743
Gross profit	\$ 264,757
Selling expense	\$ 146,250
Distribution center operating expense	\$ 182,000
Total selling expense	\$ 328,250
Distribution center depreciation expense	\$ 41,667
Idle factory expense	\$ 0
Other operating expenses	\$ 41,667
Operating income	\$ (105,160)
Interest revenue	\$ 11,913
Net interest and fees revenue (expense)	\$ 11,913
Total net income before tax	\$ (93,247)
Income tax expense	\$ (32,636)
Net income after tax and before extraordinary items	\$ (60,611)
Net income	\$ (60,611)

Integrated Strategic Management

Creating Value in Complex
and Ambiguous Settings

Building Products

Building New Products

Steps

1. Do research to decide which product to make
2. Design product
- 3. Run a focus group to help determine WTP**
4. Start development
5. Determine price
6. Set up shipping agreements (to your own DC and to others if selling to others)

Building New Products

Today's Markets

- MD Heart
 - You start with a product for this market, don't make another one for the same market
- Clinical Fertility
- MD Fertility (Estrogen)
- MD Cancer (Breast)
- Law (Narcotic)

Research

- Market Research Report for the Production Game
 - Low budget research
 - Broad, not precise
- Focus Groups
 - \$20,000 per focus group
 - 10 participants per group
 - Summary metrics: Max, Med
 - All WTP included

Market Overview

More markets will open in future games

Market Overviews

Clinical

The clinical market segments are similar to the medical segments described below, except that a clinician would provide the monitor to a patient, but the patient, instead of a physician, would have primary responsibility for monitoring the data. There are two clinical market segments: cardiovascular and fertility.

Law

For these markets, a court order would require an offender to wear the monitor, continually transmitting toxicology and location data to law enforcement, generally over the course of several days or weeks. Police departments and other government organizations would purchase the monitors to measure alcohol or different families of drugs in the bloodstream of individuals, mostly parolees. These markets are made up of six independent segments. In each segment, the agencies care about a different type of toxicology monitoring, but they all care about tracking the location of the monitored individuals, and the ability to receive continuous data updates over a cellular network, as opposed to Bluetooth, connection (with rough equivalence between the 2.4 GHz and 5 GHz options).

Medical

These markets consist of physicians monitoring patients who are undergoing various therapies. There are 14 independent market segments made up of different medical practices. A department or physician would purchase a monitor appropriate for their specific practice, provide it to a patient as part of that patient's treatment, and then review the collected data online. Selling a monitor to one segment would have no effect on the consumers in the other medical segments.

Feature Descriptions

All product designs need these 5 features (GPS can be “No GPS”)

Feature Descriptions

Platform

The electronics and monitors are added to a wearable “platform” that keeps the monitors close to the skin. People that wear the monitors prefer smaller monitors (wrist bands) over larger monitors that would typically be worn under clothes. Some markets (e.g., general population, fashion-conscious consumers) have a stronger preference for this than others.

GPS

Adding a GPS to monitors allows the monitor results to be correlated with location. This feature is imperative for some markets (e.g., law enforcement), and nice-to-have for others.

Network

To be useful the monitor must be able to transmit data for storage on another device like a smartphone or a device on the cloud. Bluetooth transmits data to a smartphone or other hand-held device. The 2.4 GHz and 5 GHz interfaces allow the monitor to transmit data over cellular networks. As with GPS, this is a very important requirement for certain markets.

Power

The monitor must have power to work, and it is designed to run on a rechargeable battery. The nickel cadmium (Ni-Cd) battery is small but basic; it can only last a few hours on full charge. The polymer battery, while of similar size to the Ni-Cd, is lighter and can last several days on one charge. The two battery pack power options simply combine multiple batteries (of either the nickel cadmium or polymer variety, depending on the type of pack) into an arrangement that is wearable around the waist, which while rather bulky and heavy for the casual user, has the merit of increasing the time between charges by a factor of about ten.

Finish

The original monitor encapsulates the hardware in utilitarian plastic, which is functional, but not particularly appealing. The monitor could be encapsulated in dyed or painted plastic, with matching materials for the platform garment. Current customers have not requested new finishes but the sales representatives hope that a more fashionable looking monitor might drive downstream demand, particularly in athletic markets.

Research and Design Guide

Example: Heart View

Features

♥ Heartbeat	:	♥ Temporal	○ Blood vessel motion	:	-
💧 Dissolved gasses	:	-	△ Toxicology	:	-
H Hormone	:	-	* Metabolic	:	-
C Cancer	:	-	🗨 Neurotoxins	:	-
🏃 Motion	:	-	🧑 Platform	:	👤 chest
📍 GPS	:	📍 GPS	📶 Networking	:	📶 2.4 GHz
🔋 Power	:	🔋 Ni-Cd battery	☐ Finish	:	Original

How many days to develop Heart View?

What is Heart View's development cost?

What is Heart View's materials cost/unit?

Attribute	Feature	Development Days	Development Cost	Materials Cost per Unit
Heartbeat	None	3	1,000	0
	Pulse only	15	30,000	15
	Temporal	90	135,000	25
Platform	Wrists	90	135,000	20
	Chest	15	3,000	10
	Sleeves	30	30,000	15
	Stockings	30	30,000	15
GPS	No GPS	3	1,000	0
	GPS	30	45,000	50
Network	Bluetooth	15	1,000	5
	2.4 GHz	30	30,000	10
	5 GHz	45	36,000	20
Power	Ni-Cd	5	1,500	5
	Ni-Cd pack	10	15,000	20
	Polymer	5	1,500	35
	Polymer pack	10	15,000	140
Finish	Original	3	2,400	0
	Blue	5	3,000	3
	Red	5	3,000	3
	Green	5	3,000	3
	Black	5	3,000	3
	White	5	3,000	3
	Metallic	90	27,000	6
	Geometric	90	27,000	6
	Camouflage	20	27,000	6

Market Research Report for Production Game

MD Heart Market

MD Heart Market

This market is comprised mainly of cardiovascular physicians who provide monitors to patients concerned with heart disease. The temporal monitor might be a convenient replacement for on-site evaluations that cost \$250-\$500.

Regional Market Size

Region	Market Size
Each Region	20,000-40,000

Bass Model Parameters

Parameter	Value
p	0.0002
incremental p from every \$500 in daily advertising	0.0002
q	0.0035

Note: incremental p from advertising only applies to the product on which it is spent, in the distribution center where it is spent, and on the day that it is spent.

Willingness to Pay (WTP)

Core Feature (Heartbeat)	WTP
Pulse only	0-115
Temporal	600-865

Note: These WTP values assume products that contain a reasonable set of features beyond one of the core features listed in the table.

Deal Breakers

- While not a complete deal breaker for all customers, lack of GPS can significantly drive down the perceived value of the product, since it's seen as a safety mechanism for dealing with patients who might experience an acute cardiovascular episode.

Other Considerations

- Minor considerations such as the convenience of various power configurations and the appeal of upgraded product finishes may also affect WTP.
- Average Days Sales Outstanding (DSO) = 30.

Market Research Report for Production Game

Clinical Fertility Market

Clinical Fertility Market

The fertility B2C market is made up of women who are interested in continually monitoring their fertility cycles. They currently can buy a device that will monitor their Reproductive LH at a cost of \$50-\$150. Your company is capable of creating a more convenient version of this device, as well as a combined LH / FSH device that will increase the monitor's precision.

Regional Market Size

Region	Market Size
Each Region	40,000-60,000

Bass Model Parameters

Parameter	Value
p	0.00025
incremental p from every \$500 in daily advertising	0.00025
q	0.004

Note: incremental p from advertising only applies to the product on which it is spent, in the distribution center where it is spent, and on the day that it is spent.

Willingness to Pay (WTP)

Core Feature (Hormone)	WTP
LH	130-300
LH / FSH	230-400

Note: These WTP values assume products that contain a reasonable set of features beyond one of the core features listed in the table.

Deal Breakers

- The discomfort of wearing bulky battery packs.

Other Considerations

- The end users in this market have a slight preference for wearing devices on their wrists and rather dislike wearing them around their chests.
- Minor considerations regarding GPS connectivity and upgraded product finishes may also affect WTP.
- Average Days Sales Outstanding (DSO) = 10.

Design Product

- Go to your HQ
- Click on the green check next to designs in the marketing panel

 **Marketing**

Designs : 0 ... 0  1  0 

Retail sales : 1 of 50 regional markets
 1 inventories
 \$0 per day advertising
 2,837 units sold total
 \$1,985,900 total revenues

Sales to other teams : 0 inventories
 0 units sold total
 \$0 total revenues

Westeros : Product Configuration [-] [x]

Product8 configuration As of day 364.51 ↻

Parent Product: Heart View 

New Product Name: ✖

Development Time 158.00 days Development Cost \$ 214,000.00 Materials cost \$ 138.00 per unit

Features

♥ Heartbeat	: None [3 days,] ✖	○ Blood vessel motion	: - 
💧 Dissolved gasses	: - 	△ Toxicology	: - 
H Hormone	: LH and FSH [60] ✖	✱ Metabolic	: - 
⊖ Cancer	: - 	🦠 Neurotoxins	: - 
🏃 Motion	: - 	🧑 Platform	: wrist [90 days,] ✖
📍 GPS	: GPS 	📶 Networking	: 2.4 GHz 
⚡ Power	: Ni-Cd battery 	☐ Finish	: Green [5 days,] ✖

Comment : 

Save Start Development

- Click on your Heart View product as it is the only one you have
- Click on new derivative
- **Give the product a name**
- Remove or change the features that you want to remove or change
- Add the features that you want to add
- Save
- Start development or run a focus group to help you determine whether you designed a product people will buy and get an estimate of how much they are willing to pay for it

Focus Group

- Click on Add New + 10
- Pick the correct market from the drop-down menu and click Start
- It will cost \$20,000 to run the focus group and will take 7 game days to complete it
- Click Start, Confirm

Add New Focus Group

New focus group with 10 participants over 7 days at immediate cost of \$ 20,000.

Region: Westeros

Design: Westeros-Baby Boomer

Market: Clinic - Fertility

Heartbeat: None

Dissolved gasses: None

Hormone: LH and FSH

Cancer: None

Motion: None

GPS: GPS

Power: Ni-Cd battery

Blood vessel motion: None

Toxicology: None

Metabolic: None

Neurotoxins: None

Platform: wrist

Networking: 2.4 GHz

Finish: Green

Start Close

- Your results will show in the focus group report from the drop-down menu on the Map page and in other places. We recommend going back to where you started the focus group and looking there as you will see more information than you'll see in the drop-down menu, so go to HQ, green check, click not started, click on product...

Focus group results

10 participants ☒ 100% would buy median wtp : \$317.48 [ADD NEW + 10](#)

Best willingness to pay across all focus groups : \$346.48

Start Development

- If you are happy with your product, click on Start Development.
- If you are not happy with your product, then click on a pencil and edit what you want.
- This product will take 158 days, with a development cost of \$214,000 and materials cost of \$138 per unit.

R&D cost
affects EBIT.

Westeros : Product Configuration

Baby Boomer configuration

As of day 492.47

Revision1

Created byadmin on team Westeros on day 373.19

Last modified by-

Parent ProductHeart View

New Product NameBaby Boomer

Development Time158 days

Development Cost\$ 214,000

Materials cost\$ 138 per unit

Features

Heartbeat

:

-

Dissolved gasses

:

-

Hormone

:

LH and FSH

Cancer

:

-

Motion

:

-

GPS

:

GPS

Power

:

Ni-Cd battery

Blood vessel motion

:

-

Toxicology

:

-

Metabolic

:

-

Neurotoxins

:

-

Platform

:

wrist

Networking

:

2.4 GHz

Finish

:

Green

Focus group results

10 participants

100% would buy

median wtp : \$317.48

ADD NEW + 10

Best willingness to pay across all focus groups : \$346.48

Comment :

N/A

Status :

Development Not Started

Save

Start Development

Product in Development

Now what?

Comment : N/A
Status : In Development **(Available on day 675.65)**

- Make sure to set up a shipping agreement from your factory to your DC.
- Can start to sell to other teams – set up shipping agreements and it will ship once it is produced. You can sell to them direct from the factory or from your DC.
- If selling from your factory, start a shipping agreement from your factory.
- If selling from your DC, start a shipping agreement from your DC.
- You can view your product lines in the product line architecture report form the drop-down menu.
- You can also see it in the product designs report.

Product line architecture for team Westeros

Click name to open panel.

Product Designs for team : World 1 : Westeros

Show feature : ☒ Icon ☐ Name

Product	Owner	Status	Start date	Ready date	Design days	Design cost	Materials unit cost	♥	○	💧	△	H	*	C	💬	👤	📍	🌐	🔍	📦
Heart View	👤 Westeros	Ready	-	-	0	\$0	\$100	♥	-	-	-	-	-	-	-	👤	📍	🌐	🔍	📦
Baby Boomer	👤 Westeros	In Development	517.65	-	158	\$214,000	\$138	-	-	-	-	H+	-	-	-	👤	📍	🌐	🔍	📦

Heart View



\$0 and 0 days to develop

Ready on day 0

Materials cost \$100 per unit

Baby Boomer



\$214,000 and 158 days to develop

in-process

Materials cost \$138 per unit

Shipping Agreement

Set up as soon as your product is in development

New Shipping Agreement

Destination : Local DC in Hamazon [Hamazon]

Product : Macrosoft-Circulation

Revision : 1

Created by : admin on day 92

Orders

Order quantity : 100

Unit price (\$) : 400.0

Days sales outstanding : 0

Ship mode : Container

Ship time : 21 days

Cost to ship 1,000 units : \$ 10,000

Cost to ship 100 units : \$ 10,000

Franchise fees and royalties

Fees paid by Local DC in Hamazon to Local DC in Macrosoft

One time at inception (\$) : 0.00

Daily until termination (\$) : 0.00

Upon cash receipt from each retail customer : \$ 0.00 per unit plus 0.00 % of retail price.

Slotting fees and buydowns

Fees paid by Local DC in Macrosoft to Local DC in Hamazon

One time at inception (\$) : 0.00

Daily until termination (\$) : 0.00

Limits

☐ Priority :

☐ Reorder point :

☐ Retail price :

Termination and returns

May terminate at will after 0 days

Early termination fee (\$) : 0

☐ Returns allowed ?

Penalty for late shipments

0.00 % per day of order price,

starting 0 days after shipping agreement is triggered, until either a batch is shipped or discount reaches full order price.

Other terms

Other terms

Comment

Comment

Save

SAVE AND PROPOSE

Cancel

Questions to think about

- What should we charge our retail customers?
- What should we charge our wholesale customers?
- How many customers do we expect per day?
- How big should our batch sizes be?
- How much labor do we put in our factory on a daily basis?
- How big should our factory be?
- How many can we produce per day?
- Do we have excess capacity?
- How should we ship our goods?
- How long does it take for our wholesalers to receive our goods?
- How long does it take for us to receive our goods?
- What terms should we give our wholesalers?
- How do we manage our company?
- Who does what?
- Other?

Pay attention to Mail

We put tips and important announcements in there!

Teams put offers in there!

Name your products



Use Your Production Model Template!

Bench Factory Parameters	
A	0.0090
α	0.10
β	0.85
$\alpha+\beta$	0.95
Setup time, in days	0.05
Batch size	100
Minimum capital	\$ 1

$$Y = A K^{\alpha} L^{\beta}$$

Depreciation years 15

of days per year 364

Materials cost per unit \$ 100

K ↓ vs L _{daily} →	Daily Throughput in units, including setup time						
	500	1,000	2,500	3,000	4,000	8,000	16,000
100,000	2.310	4.161	9.044	10.552	13.455	24.123	43.065
200,000	2.476	4.459	9.690	11.305	14.414	25.832	46.085
300,000	2.578	4.643	10.089	11.770	15.006	26.887	47.946
400,000	2.654	4.778	10.382	12.112	15.441	27.661	49.311
500,000	2.713	4.886	10.615	12.383	15.786	28.276	50.396
600,000	2.763	4.975	10.809	12.610	16.075	28.789	51.299
700,000	2.806	5.052	10.976	12.804	16.322	29.229	52.075
800,000	2.844	5.120	11.123	12.975	16.540	29.616	52.757
900,000	2.877	5.181	11.254	13.128	16.734	29.962	53.365
1,000,000	2.908	5.235	11.372	13.266	16.910	30.275	53.915
2,000,000	3.116	5.610	12.184	14.211	18.113	32.412	57.673
3,000,000	3.245	5.841	12.685	14.795	18.855	33.731	59.988
4,000,000	3.340	6.011	13.052	15.224	19.400	34.698	61.685
5,000,000	3.415	6.147	13.345	15.564	19.833	35.467	63.033

K ↓ vs L _{daily} →	Manufacturing Overhead per unit						
	500	1,000	2,500	3,000	4,000	8,000	16,000
100,000	224.333	244.742	278.453	286.043	298.640	332.391	371.956
200,000	216.725	232.495	261.780	268.609	280.046	311.109	347.981
300,000	215.227	227.222	253.244	259.552	270.219	299.588	334.854
400,000	216.033	224.626	247.863	253.747	263.799	291.869	325.957
500,000	218.022	223.430	244.147	249.660	259.183	286.165	319.305
600,000	220.717	223.084	241.454	246.630	255.677	281.704	314.039
700,000	223.872	223.306	239.449	244.312	252.919	278.083	309.710
800,000	227.347	223.928	237.937	242.503	250.699	275.067	306.056
900,000	231.054	224.848	236.794	241.077	248.882	272.505	302.910
1,000,000	234.934	225.996	235.937	239.948	247.379	270.297	300.159
2,000,000	277.998	243.549	235.259	236.873	241.065	258.121	283.776
3,000,000	323.410	265.252	240.407	239.907	241.286	253.460	275.878
4,000,000	369.097	288.219	247.664	245.185	243.949	251.672	271.259
5,000,000	414.601	311.675	255.959	251.583	247.852	251.378	268.363

Advertising

Feel free to practice advertising...

As of day 364 

Name : Local DC in Metropolis
Owner : Metropolis
Location : Metropolis
Status : READY 
Operating income : \$ 381,983 

Total advertising : \$ 0 
Total inventory : 363 units 
On-hand inventory : 263 units 
Price : \$ 700 
Total retail sales : 2,837 units 

1 Inventory

10

▼

 records

Include Inactive: ☐ Search:

Product	\$	Units	Advertising	Last date																
Name	Price	On hand	Daily \$	Any	♥	○	△	H	*	C	💬	🔍	👤	📍	🔗	🔧	🔑	🔒	🔓	
Heart View	700	263	0	363.91	♥	-	-	-	-	-	-	-	-	👤	📍	🔗	🔧	🔑	🔒	
			✓	✗																

Showing 1 to 1 of 1 entries

<

1

>

Things to Consider

- Market sizes are the same in all regions, with the exception of Metropolis
- Willingness to pay is the same in all regions
- You can trade with all teams
- You can change your batch sizes
- You can speed up or slow down your factory
 - Through CAPEX
 - Through Daily Expenditure
- You can use containers or mail for shipping
 - Consider time
 - Consider how many units you need in a container to equal the per unit mailing cost
- You can ship to wholesale customers from the factory or from your DC
- You can only sell to retail customers through the DCs

**Use the Quick Reference
Guide on Canvas!**

Things to Remember

Seriously, read these!

- You can build new factories.
- You can build new distribution centers.
- Set up shipping agreements at the same time you start construction on factories and DCs, and at the same you start development of new products.
- Name your factories and products in a way that makes them easy to identify.
- You can speed up or slow down your factory. Never set labor to a very low amount. Set it either to 0 to shut down the factory or to something reasonable for production. Only add incremental capital to a factory.
- Understand why factories go idle and what to do about it. Understand why you stock out and what to do about it. Watch your inventory status report to identify service interruptions and build up of inventory.
- Make sure the products you design or purchase include the features you think they do.
- Make sure you run focus groups for the correct markets.
- Don't sell things for less than what they cost you to produce.
- You can trade with anyone. Agree on terms through chat or in person before sending shipping agreements.
- When sending shipping agreements to other teams, include some WTP information in the comment section to cut down on negotiation time.
- Remember that if you find a market in your region, the same market, WTP and market size exists in all other regions.
- Don't forget that the commission and handling charges are paid by the retailer and shipping is paid by the wholesaler. Incorporate that information in your offers.
- You can use containers or mail for shipping.
- You can ship to wholesale customers from your DC or your factory.
- You can only sell to retail customers through the DCs.
- Make sure you understand how priorities and reorder points work. Make your reorder points > 0 for active shipping agreements. To suspend a shipping agreement, put the re-order point at -1 .

Things to Remember

Seriously, read these!

- Use the scheduling features to shut down factories, start and stop advertising, schedule construction projects, start development, among other things.
- Review your financials frequently to catch problems.
- Make sure the left hand knows what the right hand is doing.
- Choose strategic partners wisely.
- Do not use outdated course materials; only use those for today.
- Use the Quick Reference guide.
- Use the Player Manual.
- Use the reports.
- Use your models. We made them for a reason.
- Watch the chat panel as we put tips and answer questions there, and other teams may be trying to contact you.

Goal

SUPPLY = DEMAND

Integrated Strategic Management

Creating Value in Complex
and Ambiguous Settings

Next Steps

Next Steps

- Tonight: Assignment 3 and the Production Game
- Tomorrow: Practice Game

Let's Play!